

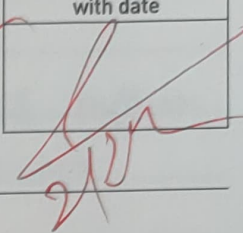
Name: Arjan. D. More Class: Sec-9 Batch: A

Subject: PL Roll No.: 12 Exp. No.: _____

Name of the Experiment: assignment 1

Performed on: _____

Submitted on: _____

Marks	Teacher's Signature with date
10	

Q1 List the Key features and advantages of python. Also mention any two Python IDEs or editors used for program development

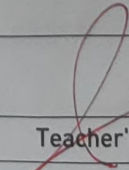
→ Features of python are:-

- Simple and easy to learn syntax
- Interpreted and high-level language
- Object-oriented and modular
- Open source and free

Advantages

- Faster development time.
- Easy debugging and maintenance
- Strong community support
- Used in web development, AI, ML, data science

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐


Teacher's Signature

Python IDE's / Edition

1] Pycharm

2] IDLE

Q2 Explain the history and Evolution of python
Describe why python has become popular
for beginner and industry application

- Python was developed by Guido van Rossum in the late 1980's and released in 1991. It was designed to complogue code readability and simplicity Evolution:
- Python 1.2 - Basic feature
- Python 2.2 - Enhanced performance and libraries
- Python 3.2 - Better Syntax, Unicode, Support improved security

Why python is popular

- Easy for beginners
- English like syntax
- Widely used in industry
- Supports multiple programming paradigms

Q3 Write a python program to accept two numeric values from the user and perform

the following operations : Addition, subtraction, multiplication, Division. Display the result Using proper input and output prompting

```
→ a = float(input("Enter First number : "))  
b = float(input("Enter Second number : "))  
print("Addition : ", a + b)  
print("Subtraction : ", a - b)  
print("Multiply : ", a * b)  
print("Division : ", a / b)
```

Q4 Analyze the following python code and identify the data types of each variables Also explain the output.

```
1 a = 10  
2 b = 5.5  
3 c = 3+4j  
4 d = True  
5 Print f (a+b, type(c), d)
```

Data types

a → int

b → float

c → complex

d → bool

output Explanation

$a+b \rightarrow 15.5$

`type(c) → <Class 'complex'>`

`d → True`

output : python

`15.5 < Class 'complex' > True`

Evaluate the use of relational, logical and bitwise operators in python. Justify with suitable examples with each type of operator is most appropriate

→ Relational operator:

$>, <, >=, <=, ==, !=$

Used for comparison

`a=5`

`b=10`

`print(a < b)`

Logical operators

and, or, not

Used in decision making

`print(a < b and b > 5)`

Bitwise operators : $&, |, ^, \sim, <<, >>$

Used for low-level operations

code: `x=5`

`y=3`

`print(x & y)`

6 Design and important implement a menu driven python program that:

→ Accept user input.

→ Performs arithmetic, relational, logical and assignment operation

→ Display result clearly

The program should handle int, float and Boolean data types

Python

```
print ("1. Arithmetic")
```

```
print ("2. Relational")
```

```
print ("3. Logical")
```

```
print ("4. Assignment")
```

```
choice = int (input ("Enter choice:"))
```

```
a = int (input ("Enter first value:"))
```

```
b = int (input ("Enter second value:"))
```

```
if choice == 1:
```

```
print ("Add: ", a+b)
```

```
print ("Sub: ", a-b)
```

```
print ("Mul: ", a*b)
```

```
print ("div: ", a/b)
```

```
else if choice == 2:
```

```
print (a>b, a<b, a==b)
```

```
else if choice = 3;  
    print + (a > 0 and b < 0)  
else if choice = 4;  
    a * = b  
    print ("After assignment = ", a)
```

7 Write a program in python to display "Hello world".

```
Print ("Hello world")
```

8 Average of 5 Numbers

```
→ a = float(input("Enter number 1:"))  
b = float(input("Enter number 2:"))  
c = float(input("Enter number 3:"))  
d = float(input("Enter number 4:"))  
e = float(input("Enter number 5:"))
```

```
avg = (a+b+c+d+e)/5  
print ("Average:", avg)
```

9 Write a prg that prompts the user to enter their age and prints out their age in years, months and days